

# Reaction–Diffusion Exchange Economies with Aggregated Cobb–Douglas Preferences for $n \geq 2$ Goods

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## Abstract

This document extends the two-good zero-sum exchange model originally developed in *Emergent Properties of Distributed Agents with Two-Stage Convex Zero-Sum Optimal Exchange Network* (Tristain 2024) to an economy with  $n \geq 2$  goods. The extension becomes feasible by aggregating pairwise Cobb–Douglas preference parameters into a globally consistent Cobb–Douglas exponent vector. This aggregation is valid provided a cycle consistency condition holds. The resulting  $n$ -good exchange economy preserves the convexity, zero-sum conservation, and tractability of the original model, while enabling higher-dimensional specialization, pattern formation, and reaction–diffusion dynamics analogous to those described by Turing (Turing 1952).

## 1 Introduction

In the two-good case [Tristain 2024], let  $z_{i \rightarrow j}^t \in \mathbb{R}^2$  denote the (signed) transfer vector sent from agent  $i$  to agent  $j$  at time  $t$ . Zero-sum conservation at the pairwise P2P trade level is expressed as:

$$z_{i \rightarrow j}^t + z_{j \rightarrow i}^t = \mathbf{0}$$

Equivalently, every bilateral exchange is a reallocation: no net goods are created or destroyed by the exchange step. Each trade is solved via convex optimization using a Cobb–Douglas utility function expressing convex preferences per agent  $i$  and preference parameter  $\alpha_{a \rightarrow b}^{(i)}$  for every pair of assets  $(a, b)$ :

$$u^{(i)}(x_a, x_b) = x_a^{\alpha_{a \rightarrow b}^{(i)}} x_b^{1 - \alpha_{a \rightarrow b}^{(i)}}$$

To reinterpret this system as a *reaction–diffusion economy*, we draw inspiration from Turing’s reaction–diffusion dynamics (Turing 1952). In this analogy:

- the **reaction phase** corresponds to each agent applying transformation rules (or production technologies) to its own endowment, and
- the **diffusion phase** corresponds to agents participating in structured exchange with the rest of the system.

This produces endogenous spatial or distributional patterns in the economy, just as chemical species form patterns under reaction–diffusion rules.

To generalize this framework to  $n \geq 2$  goods while preserving convexity and analytic tractability, we must generalize the utility functions. The key insight is that pairwise Cobb–Douglas preferences can be aggregated into a globally consistent Cobb–Douglas utility function, enabling the full  $n$ -good version of the reaction–diffusion exchange economy.

## 2 Pairwise Cobb–Douglas Preferences

Let the set of  $n$  goods indexes be:

$$\mathcal{N} = \{1, 2, \dots, n\}.$$

Let the set of  $m$  agent indexes be:

$$\mathcal{M} = \{1, 2, \dots, m\}.$$

For each pair  $a \neq b$ , an agent  $i \in \mathcal{M}$  specifies the pairwise Cobb–Douglas preference:

$$u_{(a,b)}^{(i)}(x_{i,a}, x_{i,b}) = x_{i,a}^{\alpha_{a \rightarrow b}^{(i)}} x_{i,b}^{1 - \alpha_{a \rightarrow b}^{(i)}}, \quad \alpha_{a \rightarrow b}^{(i)} \in (0, 1).$$

Symmetry imposes:

$$\alpha_{b \rightarrow a}^{(i)} = 1 - \alpha_{a \rightarrow b}^{(i)}.$$

### 2.1 Pairwise Ratios

From each pair index  $a, b \in \mathcal{N}$ , define:

$$r_{a \rightarrow b}^{(i)} = \frac{1 - \alpha_{a \rightarrow b}^{(i)}}{\alpha_{a \rightarrow b}^{(i)}} = \frac{\alpha_{b \rightarrow a}^{(i)}}{1 - \alpha_{b \rightarrow a}^{(i)}} > 0,$$

representing the relative intensity of preference for good  $b$  over good  $a$ .

## 3 Aggregation to Global Cobb–Douglas Utility

For every agent  $i$  we seek a global utility representation for  $n$  goods:

$$u^{(i)}(\mathbf{x}) = \prod_{k=1}^n x_{i,k}^{\beta_k^{(i)}}, \quad \beta_k^{(i)} \in [0, 1],$$

such that the pairwise marginal rates of substitution match the pairwise preferences.

### 3.1 Consistency Condition

For a fixed agent  $i$ , define ratios

$$r_{a \rightarrow b}^{(i)} = \frac{1 - \alpha_{a \rightarrow b}^{(i)}}{\alpha_{a \rightarrow b}^{(i)}}.$$

A globally consistent exponent vector  $\beta^{(i)} \in \mathbb{R}_{++}^n$  exists (unique up to scale) if and only if for every cycle of goods ( $g_1 \rightarrow g_2 \rightarrow \dots \rightarrow g_n \rightarrow g_1$ ) we have:

$$\prod_{\ell=1}^n r_{g_\ell \rightarrow g_{\ell+1}}^{(i)} = 1, \quad g_{n+1} = g_1.$$

With:

$$\beta^{(i)} = (\beta_1^{(i)}, \beta_2^{(i)}, \dots, \beta_n^{(i)}) \in \mathbb{R}_{++}^n.$$

Pairwise coefficients must satisfy:

$$\frac{\beta_a^{(i)}}{\beta_a^{(i)} + \beta_b^{(i)}} = \alpha_{i,(a,b)} \quad \forall a, b \in \mathcal{N}, a \neq b$$

Equivalently:

$$\frac{\beta_b^{(i)}}{\beta_a^{(i)}} = r_{i,(a,b)}$$

### 3.2 Constructing $\beta^{(i)}$

Choose a reference good, say  $\beta_1^{(i)} = 1$ , then recursively define:

$$\beta_j^{(i)} = r_{i,(1,j)} \beta_1^{(i)}.$$

Normalize:

$$\tilde{\beta}_k^{(i)} = \frac{\beta_k^{(i)}}{\sum_{j=1}^n \beta_j^{(i)}}.$$

This yields a unique (up to scale) global Cobb–Douglas exponent vector.

## 4 The $n$ -Good Reaction–Diffusion Exchange Economy

### 4.1 Endowments and Conservation

Each agent  $i$  holds an endowment vector in time  $t$ :

$$\mathbf{x}_i^t = (x_{i,1}^t, \dots, x_{i,n}^t) \in \mathbb{R}_+^n,$$

with global constant conservation  $\mathbf{E} \in \mathbb{R}_+^n$  for all agents  $i \in \mathcal{N}$  in all time sequences  $t \in \{1, 2, \dots, T\}$ :

$$\sum_{i=1}^k \mathbf{e}_i^t = \mathbf{E}$$

## 4.2 Reaction Phase

All agents  $i \in \mathcal{N}$  apply linear technologies  $r \in \mathcal{R}$  such that:

$$\begin{aligned}\mathbf{O}_r &\in \mathbb{R}_+^n && (\text{outputs}) \\ \mathbf{I}_r &\in \mathbb{R}_+^n && (\text{inputs}) \\ \mathbf{e}_i^{t,\text{react}} &= \mathbf{e}_i^t + \mathbf{e}_i^t(\mathbf{O}_r - \mathbf{I}_r), && \mathbf{e}_i^t \mathbf{I}_r \leq \mathbf{e}_i^t.\end{aligned}$$

with  $\mathbf{I}_r$  being the endowment of goods to be consumed (inputs) for producing  $\mathbf{O}_r$  (outputs), given the initial endowment  $\mathbf{e}_i^t$  for period  $t$ , agent  $i$  and set of technologies  $\mathcal{R}$ .

## 5 Diffusion (Exchange) Phase

Let  $\mathbf{p}^t \in \mathbb{R}_{++}^n$  be the price vector. Agent  $i$  chooses  $\lambda_i^t \in \mathbb{R}_+^n$  and a trade bundle  $\mathbf{z}_i^t \in \mathbb{R}_+^n$ :

$$\mathbf{z}_i^t = \lambda_i^t \mathbf{e}_i^{t,\text{react}}, \quad x_{i,k}^t(\lambda_i^t) = (1 - \lambda_i^t) e_{i,k}^{t,\text{react}} + z_{i,k}^t.$$

Utility:

$$u_i(\mathbf{x}_i^t) = \prod_{k=1}^n x_{i,k}^{\tilde{\beta}_k^{(i)}}.$$

Budget:

$$\mathbf{p}^t \cdot \mathbf{z}_i^t \leq \lambda_i^t (\mathbf{p}^t \cdot \mathbf{e}_i^{t,\text{react}}).$$

Demand:

$$z_{i,k}^t(\lambda_i^t) = \frac{\tilde{\beta}_k^{(i)}}{p_k^t} \lambda_i^t (\mathbf{p}^t \cdot \mathbf{e}_i^{t,\text{react}}).$$

Derivative of log-utility in  $\lambda_i^t$ :

$$\frac{d}{d\lambda_i^t} \log u_i(\lambda_i^t) = \sum_{k=1}^n \tilde{\beta}_k^{(i)} \frac{-e_{i,k}^{t,\text{react}} + \frac{\tilde{\beta}_k^{(i)}}{p_k^t} (\mathbf{p}^t \cdot \mathbf{e}_i^{t,\text{react}})}{(1 - \lambda_i^t) e_{i,k}^{t,\text{react}} + \lambda_i^t \frac{\tilde{\beta}_k^{(i)}}{p_k^t} (\mathbf{p}^t \cdot \mathbf{e}_i^{t,\text{react}})}.$$

## 6 Market Clearing

Supply:

$$Q_k^t = \sum_i \lambda_i^t e_{i,k}^{t,\text{react}}.$$

Demand:

$$D_k^t = \sum_i \frac{\tilde{\beta}_k^{(i)}}{p_k^t} \lambda_i^t (\mathbf{p}^t \cdot \mathbf{e}_i^{t,\text{react}}).$$

Clearing requires  $D_k^t = Q_k^t$ .

## 7 Conclusion

Aggregating pairwise Cobb–Douglas preferences into global preferences allows the two-good zero-sum model of Tristain 2024 to be extended to an economy with arbitrary dimension  $n$ . The resulting reaction–diffusion exchange economy maintains convexity, tractability, and conservation, while opening the door to richer emergent behaviors in higher-dimensional economic systems.

## References

- Tristain, Diego Correa (2024). *Emergent Properties of Distributed Agents with Two-Stage Convex Zero-Sum Optimal Exchange Network*. GitHub repository. [https://github.com/onedge-network/Emergent\\_Properties\\_paper](https://github.com/onedge-network/Emergent_Properties_paper).
- Turing, Alan M. (1952). “The Chemical Basis of Morphogenesis”. In: *Philosophical Transactions of the Royal Society of London. Series B* 237.641, pp. 37–72. DOI: 10.1098/rstb.1952.0012. URL: <https://doi.org/10.1098/rstb.1952.0012>.